

Comparing Dirichlet-Multinomial Parameterizations for Fitting Composition Data

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11 July 2025

This document provides notation for a study of the properties of the Dirichlet-Multinomial distribution for fitting composition data in stock assessment applications. There are two parametric forms of the Dirichlet-Multinomial distribution: a linear form and a saturated form. The overall goal of the study is to determine which Dirichlet-Multinomial distribution (DM) form performs best under various assessment modeling conditions.

We begin by specifying the notation used in the study, noting that there are at least two alternatives for the DM from Thorson et al. (2017) and Fisch et al. (2021). Our first goal is to provide a glossary of terminology to describe the data and parameters needed to fit composition data sampled from a fish stock in this study (Table 1).

Composition data are counts or proportions of individuals classified into predefined categories—such as age or size classes—capturing the distribution of characteristics within a statistical sampling frame. We will use age-structured data as the type of composition data sampled from the fish stock. The number of categories, or age classes, is denoted by K , with the age bins $k=1, 2, \dots, (K-1)$ representing true ages and the age bin K representing all individuals K and older.

We use P_k to denote the proportion of individuals in category k . There are three types of proportion vectors used in this study. The true proportion vector at age in the stock is P^{TRUE} . The observed proportion vectors in the sampling frame are P^{OBS} . The predicted proportion vectors estimated by optimizing the parameters of the DM likelihood forms fitted to the observed data are P^{PRED} . These three proportion types are for simulating observed proportion data with a known sample size from the true population and estimating the predicted proportions. The simulated data will include observation error from random sampling and can also include process error due to random year-to-year fluctuations from environmental and demographic factors.

We use X_k to denote the count of individuals in category k . As with the proportion types, there are three types of count vectors used in this study. The true vector of absolute numbers at age in the stock is X^{TRUE} . The observed count vectors from the sampling frame are X^{OBS} . The predicted count vectors estimated by optimizing the parameters of the DM likelihood forms fitted to the observed data are X^{PRED} . Typically, the

absolute and predicted counts of individuals are modeled as continuous variables for computational tractability while the observed composition data are discrete counts of individuals.

Table 1. Glossary of terms.

Variable	Description
K	Number of categories of age or size bins indexed by $k = 1, 2, \dots, K$
$\underline{P}_y^{TRUE}$	True population proportion vector in year y , with elements $P_{y,k}^{TRUE}$
$\underline{P}_y^{OBS}$	Observed population proportion vector in year y , with elements $P_{y,k}^{OBS}$
$\underline{P}_y^{PRED}$	Predicted population proportion vector in year y , with elements $P_{y,k}^{PRED}$
$\underline{X}_y$	Absolute population numbers at age vector in year y , with elements $X_{y,k}$
$\underline{X}_y^{OBS}$	Observed population count vector in year y , with elements $X_{y,k}^{OBS}$
$\underline{X}_y^{PRED}$	Predicted population count vector in year y , with elements $X_{y,k}^{PRED}$
$n_{input,y}$	Observed sample size for the observed proportions in year y
$\pi_{y,c}$	Predicted proportion for category c in year y
θ	Weighting parameter for the linear Dirichlet-multinomial distribution
β	Weighting parameter for the saturating Dirichlet-multinomial distribution
$n_{eff,y}$	Effective sample size for the observed proportions in year y

Here is the multinomial NLL for a single time period, where underlining indicates a vector with dimension number of categories. The vector of estimated, or predicted proportions $\underline{\pi} = (\pi_1, \dots, \pi_C)$, is a function of the input sample size n_{input} and the observed proportions $\underline{p} = (p_1, \dots, p_C)$.

$$(1.1) \quad NLL_{MN}(\underline{\pi} | n_{input}, \underline{p}) = -n_{input} \sum_c p_c n_{input} \cdot \ln(\pi_c)$$

Here is the multinomial NLL for a number of time periods indexed by y , where double underlining indicates a matrix with dimensions number of categories and number of time periods.

$$(1.2) \quad NLL_{MN}(\underline{\underline{\pi}} | \underline{n_{input}}, \underline{\underline{p}}) = -\sum_y n_{input,y} \sum_c p_{y,c} n_{input,y} \cdot \ln(\pi_{y,c})$$

Here is the linear Dirichlet Multinomial NLL for a single time period. The vector of predicted proportions $\underline{\pi} = (\pi_1, \dots, \pi_C)$ and the predicted linear scale parameter θ are a function of the input sample size n_{input} and the observed proportions $\underline{p} = (p_1, \dots, p_C)$.

$$(1.3) \quad \begin{aligned} NLL_{DML}(\underline{\pi}, \theta | n_{input}, \underline{p}) = & -\ln(\Gamma(n_{input} + 1)) + \sum_c \ln(\Gamma(n_{input} p_c + 1)) \\ & -\ln(\Gamma(\theta n_{input})) + \ln(\Gamma(n_{input} + \theta n_{input})) \\ & -\sum_c \ln(\Gamma(n_{input} p_c + \theta n_{input} \pi_c)) + \sum_c \ln(\Gamma(\theta n_{input} \pi_c)) \end{aligned}$$

Here is the linear Dirichlet Multinomial NLL for a set of time periods indexed by y .

$$(1.4) \quad NLL_{DML}(\underline{\underline{\pi}}, \theta | \underline{n_{input}}, \underline{\underline{p}}) = \sum_y \left\{ \begin{aligned} & -\ln(\Gamma(n_{input,y} + 1)) + \sum_c \ln(\Gamma(n_{input,y} p_{y,c} + 1)) \\ & -\ln(\Gamma(n_{input,y} \theta)) + \ln(\Gamma(n_{input,y} \theta + n_{input,y})) \\ & -\sum_c \ln(\Gamma(n_{input,y} p_{y,c} + \theta n_{input,y} \pi_{y,c})) + \sum_c \ln(\Gamma(\theta n_{input,y} \pi_{y,c})) \end{aligned} \right\}$$

Here is the saturating Dirichlet Multinomial NLL for a single time period. The vector of predicted proportions $\underline{\pi} = (\pi_1, \dots, \pi_C)$ and the predicted saturation parameter β are a function of the input sample size n_{input} and the observed proportions $\underline{p} = (p_1, \dots, p_C)$.

$$\begin{aligned}
(1.5) \quad NLL_{DMS}(\underline{\pi}, \beta | n_{input}, \underline{p}) = & -\ln(\Gamma(n_{input} + 1)) + \sum_c \ln(\Gamma(n_{input} p_c + 1)) \\
& -\ln(\Gamma(\beta)) + \ln(\Gamma(\beta + n_{input})) \\
& -\sum_c \ln(\Gamma(n_{input} p_c + \beta \pi_c)) + \sum_c \ln(\Gamma(\beta \pi_c))
\end{aligned}$$

Here is the saturating Dirichlet Multinomial NLL for a set of time periods indexed by y .

$$(1.6) \quad NLL_{DMS}(\underline{\pi}, \beta | \underline{n_{input}}, \underline{p}) = \sum_y \left\{ \begin{aligned} & -\ln(\Gamma(n_{input,y} + 1)) + \sum_c \ln(\Gamma(n_{input,y} p_{y,c} + 1)) \\ & -\ln(\Gamma(\beta)) + \ln(\Gamma(\beta + n_{input,y})) \\ & -\sum_c \ln(\Gamma(n_{input,y} p_{y,c} + \beta \pi_{y,c})) + \sum_c \ln(\Gamma(\beta \pi_{y,c})) \end{aligned} \right\}$$

The effective sample size n_{eff} for the linear Dirichlet Multinomial in a single period depends on θ and n_{input} as

$$(1.7) \quad n_{eff} = \frac{1 + \theta n_{input}}{1 + \theta}$$

And for a set of periods n_{eff} is

$$(1.8) \quad n_{eff} = \sum_y n_{eff,y} = \sum_y \frac{1 + \theta n_{input,y}}{1 + \theta}$$

Similarly, the effective sample size n_{eff} for the saturated Dirichlet Multinomial in a single period depends on β and n_{input} as

$$(1.9) \quad n_{eff} = \frac{n_{input} + \beta n_{input}}{n_{input} + \beta}$$

And for a set of periods n_{eff} is

$$(1.10) \quad n_{eff} = \sum_y n_{eff,y} = \sum_y \frac{n_{input,y} + \beta n_{input,y}}{n_{input,y} + \beta}$$

First order partial derivatives for the effective sample size n_{eff} of the linear Dirichlet Multinomial as a function of the input parameters are:

$$(1.11) \quad \frac{\partial n_{eff}}{\partial n_{input}} = \frac{\theta}{1 + \theta} = \frac{\theta}{1 + \theta n_{input}} n_{eff}$$

And

$$(1.12) \quad \frac{\partial n_{eff}}{\partial \theta} = \frac{n_{input} - 1}{(1 + \theta)^2} = \frac{n_{input} - 1}{(1 + \theta)(1 + \theta n_{input})} n_{eff}$$

Similarly, first order partial derivatives for the effective sample size n_{eff} of the saturated Dirichlet Multinomial as a function of the input parameters are:

$$(1.13) \quad \frac{\partial n_{eff}}{\partial n_{input}} = \frac{\beta(1 + \beta)}{(n_{input} + \beta)^2} = \frac{\beta}{n_{input}(n_{input} + \beta)} n_{eff}$$

And

$$(1.14) \quad \frac{\partial n_{eff}}{\partial \beta} = \frac{n_{input}(n_{input} - 1)}{(n_{input} + \beta)^2} = \frac{(n_{input} - 1)}{(1 + \beta)(n_{input} + \beta)} n_{eff}$$